ICAR - CENTRAL INSTITUTE FOR COTTON RESEARCH (CICR)

National Agricultural Advisory Bulletin: Comprehensive Cotton Health & Pest Management

1. Cotton Leaf Yellowing and Chlorosis Management

Leaf yellowing (chlorosis) in cotton is primarily caused by micronutrient deficiency, specifically Magnesium deficiency or Nitrogen deficiency, often exacerbated by waterlogging. When veins remain green while interveinal areas turn pale yellow or reddish (known locally as Lalio in Gujarat), foliar nutrition is urgent. Recommended Advisory: Spray Magnesium Sulphate (MgSO4) @ 10 grams per liter of water (150 grams per 15-liter pump) along with Urea @ 10-15 grams per liter. Apply 2 sprays at 15-day intervals during flowering and boll development.

2. Pink Bollworm (Pectinophora gossypiella) Control Strategy

Pink bollworm is a destructive pest causing rosetted flowers and locule damage. Monitoring must begin 45 days after sowing. Install pheromone traps @ 5 traps per acre with Gossyplure lures. If trap catches exceed 8 moths per night for 3 consecutive days or 10% rosette flowers are observed, initiate chemical intervention. Recommended Spray: Emamectin Benzoate 5% SG @ 9-10 grams per 15-liter spray pump (200 g/ha) OR Chlorantraniliprole 18.5% SC @ 6-7 ml per 15-liter spray pump. Avoid continuous use of synthetic pyrethroids to prevent pest resistance.

3. Sucking Pests (Aphids, Jassids, and Thrips)

Sucking pests infest the undersides of cotton leaves, draining sap and causing downward leaf curling, yellowing, and stunted growth. Bio-control Action: Spray commercial Neem Seed Kernel Extract (NSKE 5%) or Neem Oil (1500 PPM) @ 45-50 ml per 15-liter spray pump. Chemical Intervention: If pest population exceeds Economic Threshold Level (ETL) of 5 nymphs per leaf, spray Imidacloprid 17.8% SL @ 4-5 ml per 15-liter spray pump or Thiamethoxam 25% WG @ 6-8 grams per 15-liter pump.

4. Agro-Chemical Safety and Personal Protective Equipment (PPE)

Farmer Safety Protocol: Always wear personal protective equipment including face mask, safety goggles, rubber gloves, and protective clothing. Always spray in the direction of the wind; never spray against wind direction. Spray strictly during cool hours (morning before 10 AM or evening after 4 PM). Strictly avoid banned compounds including Endosulfan and Monocrotophos. Maintain a mandatory 15-day pre-harvest interval (PHI).